

CAPSTONE PROJECT / ENTERPRISE AI RAG SUPPORT

SCHOOLTRUST AI

Verified Parent Support and Knowledge Improvement System

Requirements-only specification for the assessed student implementation.

Project title	SchoolTrust AI
Student name	Moeen Ahmad Butt
Registration ID	[ENTER REGISTRATION ID]
Case study	Umeed Education System, Lahore
Version / date	Version 1.0 14 July 2026
Live demo link	[ENTER AFTER DEPLOYMENT]
GitHub repository	[ENTER IF AVAILABLE]

SOFTWARE REQUIREMENTS SPECIFICATION

One focused project. Approved sources. Human-controlled improvement.

Document Control

Item	Requirement
Document purpose	Define the functional requirements, user roles, use cases, business rules and external interfaces for SchoolTrust AI.
Selected capstone	Enterprise AI RAG Customer Support Platform.
Implementation boundary	One school, one website assistant, one administrator dashboard and one n8n automation project.
Requirement language	The word shall indicates a mandatory requirement. Should indicates a desirable requirement.
Prototype status	Student capstone prototype requiring owner credentials and approved school content before live use.

System Boundary

In scope	Out of scope
Parent questions about approved public school information	Private student academic, attendance or fee-ledger records
English and Urdu question handling	Voice calls and social-media omnichannel deployment
Current-source retrieval and answer guardrails	Independent AI authority to create or approve school policy
Knowledge-gap recording and administrator approval	Automatic publication without human review
Regression evaluation of critical questions	Production certification, formal penetration testing or legal compliance certification

Core requirement

A missing answer is not treated as a chatbot failure. It becomes a controlled knowledge-gap record for administrator review. The system shall prefer a safe escalation over an unsupported school fact.

1. Functional Requirements

ID	Requirement	Priority	Verification
FR-01	The system shall accept a parent question through the website webhook using JSON.	Must	API test
FR-02	The system shall reject questions shorter than 3 characters or longer than 1,000 characters.	Must	Boundary test
FR-03	The system shall detect whether the question is primarily English or Urdu.	Must	Language test

ID	Requirement	Priority	Verification
FR-04	The system shall classify the question into a supported topic such as fees, admissions, timings, contact, transport or general.	Must	Topic test
FR-05	The system shall mark fees, dates, timings, admissions, policies, payments, results and transport as critical topics.	Must	Rule test
FR-06	The system shall create a semantic embedding for every valid question.	Must	Execution log
FR-07	The system shall retrieve only approved knowledge with a non-expired effective window.	Must	Database test
FR-08	The system shall provide the language model with the question and retrieved approved sources only.	Must	Prompt inspection
FR-09	The generated response shall contain an answer mode, answer text, confidence score, source identifiers and an internal reason.	Must	Schema test
FR-10	The system shall verify that cited source identifiers exist in the retrieved source set.	Must	Guardrail test
FR-11	The system shall require a minimum confidence of 0.82 for a critical-topic verified answer.	Must	Threshold test
FR-12	The system shall require a minimum confidence of 0.72 for a non-critical verified answer.	Must	Threshold test
FR-13	The system shall request clarification when multiple valid student categories or cases require different answers.	Must	Scenario test
FR-14	The system shall safely escalate when approved information is absent, incomplete, conflicting or below the required threshold.	Must	Scenario test
FR-15	The system shall record repeated unsupported questions as grouped knowledge gaps with occurrence counts.	Must	Database test
FR-16	The system shall assign a higher initial priority to gaps involving critical topics.	Must	Rule test

ID	Requirement	Priority	Verification
FR-17	The system shall log verified, clarification and escalated conversations with topic, confidence and source identifiers.	Must	Log inspection
FR-18	The administrator dashboard shall display approved knowledge, open gaps, verified answers, escalations and seven-day test pass rate.	Must	UI/API test
FR-19	The system shall require an administrator token for dashboard and approval endpoints in the prototype.	Must	Authorization test
FR-20	An administrator shall be able to publish approved knowledge with title, category, content, version and effective dates.	Must	Approval test
FR-21	Publishing approved knowledge shall create its embedding and store an approval audit record.	Must	Database test
FR-22	Publishing an answer for a recorded gap shall resolve the related gap.	Must	Workflow test
FR-23	The system shall execute active regression questions manually and once per day.	Must	Schedule test
FR-24	The evaluation module shall store the generated answer, pass status, score and missing expected facts.	Must	Evaluation test
FR-25	The web interface shall provide suggested questions and visibly distinguish verified, clarification and escalated responses.	Should	UI inspection
FR-26	The web interface shall be responsive on desktop, tablet and mobile widths.	Should	Responsive test

2. User Roles

Role	Permissions	Restrictions
Parent / visitor	Ask public school-information questions; receive verified answers, clarification requests or escalation notices.	Cannot view the administrator dashboard, approve knowledge or access private student records.
School administrator	View knowledge health; review gaps; approve knowledge; set effective dates; resolve gaps; view test results.	Cannot expose service-role or AI keys to the browser. Must approve only authorised school information.

Role	Permissions	Restrictions
Automation scheduler	Run daily regression evaluation and store results.	Cannot approve knowledge or alter policy. Operates only on active test cases and approved sources.

3. Use Cases

UC-01 - Receive a verified answer

Field	Specification
Primary actor	Parent / visitor
Purpose	Obtain an answer supported by current approved school knowledge.
Preconditions	The workflow is active and relevant approved knowledge has an embedding.
Trigger	The parent submits a valid question.
Main flow	1. Validate the question. 2. Detect language, topic and criticality. 3. Create the embedding. 4. Retrieve approved active sources. 5. Generate a controlled response. 6. Apply source and confidence guardrails. 7. Log and return the verified answer.
Alternative flows	- If the source set is empty, continue to UC-03. - If the answer does not cite a retrieved source, continue to UC-03.
Postconditions	A verified conversation log exists and the parent receives the answer.
Related requirements	FR-01, FR-03, FR-04, FR-06, FR-07, FR-09, FR-10, FR-17

UC-02 - Request parent clarification

Field	Specification
Primary actor	Parent / visitor
Purpose	Resolve ambiguity without selecting an unsupported category.
Preconditions	Approved sources contain more than one applicable case.
Trigger	The question lacks a required category, such as new admission versus promoted student.
Main flow	1. Retrieve both applicable cases. 2. Generate a precise clarification question or state both categories. 3. Log the clarification response. 4. Return it to the parent.
Alternative flows	- If the sources conflict rather than describe valid categories, continue to UC-03.

Field	Specification
Postconditions	The system waits for a more specific parent question.
Related requirements	FR-13, FR-17

UC-03 - Escalate and record a knowledge gap

Field	Specification
Primary actor	Parent / visitor
Purpose	Avoid inventing school information and create a reviewable improvement task.
Preconditions	The question is valid but cannot be answered safely.
Trigger	No approved source, low confidence, conflict or incomplete information is detected.
Main flow	1. Return a safe parent-facing notice. 2. Normalise the topic. 3. Create or increment the matching open gap. 4. Assign priority. 5. Log the escalated conversation.
Alternative flows	- If a similar open gap already exists, increment its occurrence count instead of creating a duplicate.
Postconditions	The parent is not misled and an administrator can review the gap.
Related requirements	FR-14, FR-15, FR-16, FR-17

UC-04 - View knowledge health

Field	Specification
Primary actor	School administrator
Purpose	Review the operational state of the assistant.
Preconditions	A valid prototype administrator token is supplied.
Trigger	The administrator loads dashboard data.
Main flow	1. Validate the token. 2. Calculate knowledge and conversation metrics. 3. Load open gaps and recent test results. 4. Return the dashboard response. 5. Render live metrics.
Alternative flows	- Reject an absent or invalid token.
Postconditions	No data is changed.
Related requirements	FR-18, FR-19

UC-05 - Approve and publish knowledge

Field	Specification
Primary actor	School administrator
Purpose	Convert reviewed school information into an approved searchable source.
Preconditions	The administrator has authoritative content and a valid token.
Trigger	The approval form is submitted.
Main flow	1. Validate mandatory fields. 2. Create the content embedding. 3. Store the source as approved. 4. Record effective dates and version. 5. Resolve the linked gap when present. 6. Write an administrator audit record.
Alternative flows	- Reject content shorter than 20 characters. - Reject an invalid administrator token.
Postconditions	The approved source is available for retrieval and the change is auditable.
Related requirements	FR-19, FR-20, FR-21, FR-22

UC-06 - Run regression evaluation

Field	Specification
Primary actor	Automation scheduler or administrator
Purpose	Verify critical answers after knowledge changes.
Preconditions	Active regression tests and approved embedded knowledge exist.
Trigger	The daily schedule fires or the manual trigger is executed.
Main flow	1. Load active tests. 2. Embed each question. 3. Retrieve approved knowledge. 4. Generate an answer using only that knowledge. 5. Compare expected facts. 6. Store score, status and missing facts.
Alternative flows	- A missing approved answer produces a failed test rather than an invented answer.
Postconditions	Evaluation results are visible on the dashboard.
Related requirements	FR-23, FR-24

4. Business Rules

ID	Business rule
BR-01	Only knowledge with status approved may support a parent answer.
BR-02	A source with a future effective date or expired date shall not be retrieved.
BR-03	Critical topics shall use the 0.82 verified-answer confidence threshold.
BR-04	Non-critical topics shall use the 0.72 verified-answer confidence threshold.
BR-05	A verified answer shall cite at least one identifier from the retrieved source set.
BR-06	The assistant shall not invent fees, dates, timings, policies, addresses or availability.
BR-07	Different valid categories shall be stated separately or clarified. They shall not be collapsed into one amount.
BR-08	Conflicting or incomplete approved information shall result in escalation.
BR-09	An administrator shall approve every knowledge publication.
BR-10	The public web interface shall never receive the Supabase service-role key or AI API key.
BR-11	Private student records are outside the public assistant's scope.
BR-12	Repeated identical open gaps within the same topic shall increment occurrence count.
BR-13	Critical-topic gaps shall initially receive high priority.
BR-14	Approved content shall contain at least 20 characters.
BR-15	The question length shall be between 3 and 1,000 characters.
BR-16	The administrator demo token is a prototype control and shall be replaced by authenticated role-based access before production use.
BR-17	The system is a controlled knowledge workflow, not a self-training model.

5. External Interfaces

5.1 User Interfaces

Interface	Requirement
Parent assistant	Responsive chat view with question input, suggested questions, conversation stream and visible response status.
Administrator dashboard	Metrics for approved knowledge, open gaps, verified answers and test pass rate; gap review; test results.

Interface	Requirement
Approval dialog	Fields for gap identifier, document title, category, approved content, effective dates and administrator name.

5.2 HTTP Interfaces

Method / path	Request	Response / control
POST /webhook/schooltrust-ask	question, session_id, optional user_contact	Verified, clarification or escalated answer. Public prototype endpoint.
GET /webhook/schooltrust-admin-data	Header: x-admin-token	Dashboard metrics, gaps and evaluations. Administrator token required.
POST /webhook/schooltrust-admin-approve	Header: x-admin-token plus approved source fields	Published knowledge identifier and approval confirmation.

5.3 Data and AI Interfaces

External system	Purpose	Required configuration
Supabase PostgreSQL with pgvector	Approved knowledge, embeddings, conversation logs, gaps, regression tests, evaluation results and audit actions.	SUPABASE_URL and server-side service-role key.
OpenAI API	Question and content embeddings; controlled answer generation; regression answer generation.	OPENAI_API_KEY, chat model and embedding model.
n8n	Webhook handling, branching, guardrails, database calls, approval automation and scheduled tests.	Imported workflow plus protected server-side configuration.
Static web host / school website	Parent assistant and administrator prototype interface.	Production n8n base URL and non-secret demo token only.

5.4 Data Contracts

Object	Required fields
Parent question	question: string; session_id: string; user_contact: optional string
Controlled answer	success: boolean; status: verified clarification escalated; answer: string; session_id: string
Approved knowledge	document_title; category; content; version_no; status; effective_from; expires_at; embedding
Knowledge gap	question; normalized_topic; reason; occurrence_count; priority; status; timestamps
Evaluation result	regression_test_id; generated_answer; passed; score; missing_facts; notes

6. Requirements Traceability

Capability	Functional requirements	Use case	Implemented workflow nodes
Verified parent answer	FR-01 to FR-12	UC-01	Parent Question Webhook through Return Verified Answer
Clarification	FR-13, FR-17	UC-02	Needs Clarification, Log Clarification, Return Clarification
Safe escalation and gap	FR-14 to FR-17	UC-03	Record Knowledge Gap, Log Escalated Conversation
Knowledge dashboard	FR-18, FR-19	UC-04	Administrator Dashboard Webhook, Load Dashboard Data
Administrator approval	FR-19 to FR-22	UC-05	Administrator Approval Webhook through Log Administrator Action
Regression evaluation	FR-23, FR-24	UC-06	Daily / Manual Quality Test through Store Evaluation Result
Responsive interface	FR-25, FR-26	UC-01, UC-04, UC-05	web/index.html, web/styles.css, web/app.js

7. Acceptance Requirements

ID	Acceptance condition
AR-01	The annual-fund question returns both new-admission and promoted-student categories or asks a precise clarification.
AR-02	A transport-fee question with no approved source returns a safe escalation and creates or increments a gap.
AR-03	An invalid administrator token cannot load dashboard data or publish knowledge.
AR-04	An administrator-approved source becomes retrievable only after its embedding is stored.
AR-05	The daily or manual evaluation branch stores a result for every active regression test.
AR-06	The web interface displays a different status label for verified, clarification and escalated responses.
AR-07	The submitted n8n JSON imports and contains the parent, approval, dashboard and testing branches.

Submission note

This SRS describes the included prototype package. Registration ID, live demo link and GitHub link remain editable because they were not provided at generation time.